Coordinates

Recall from last class:

- The subset $\{v_1, v_2, \ldots, v_p\}$ of the vector space V is **linearly independent** if

$$c_1v_1 + \cdots + c_pv_p = 0_V$$

  has only the trivial solution.

- **Theorem:** Given $\{v_1, \ldots, v_p\}$ with $p \geq 2$ and $v_1 \neq 0_V$. The vectors $v_1, \ldots, v_p$ are linearly dependent if and only if for some $j > 1$, the vector v_j is a linear combination of $v_1, \ldots, v_{j-1}$.

- Let H be a subspace of a vector space V. The set $\{v_1, \ldots, v_p\}$ is a **basis** for H if

  1. $\{v_1, \ldots, v_p\}$ is linearly independent,
  2. $H = \text{Span}\{v_1, \ldots, v_p\}$.

- **Spanning Set Theorem:** Let $S = \{v_1, \ldots, v_p\} \subseteq V$ and $H = \text{Span}\{v_1, \ldots, v_p\}$.

  1. Suppose v_k can be written as a linear combination of the other vectors in S, then $S - \{v_k\}$ still spans H.
  2. If $H \neq \{0_V\}$, some subset of S is a basis for H.

- **Theorem:** The pivot columns of an $m \times n$ matrix A form a basis for $\text{Col}(A)$.

- **Replacement Theorem:** Let V be a vector space that is generated by set G. Suppose $|G| = n$. Let L be a linearly independent subset of V with $|L| = m$. Then $m \leq n$ and there exists a subset H of G containing exactly $n - m$ vectors such that $H \cup L$ generates V.

- **Corollary:** Let V be a vector space with a finite basis, then every basis has the same number of vectors.

Theorem: Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ be a basis for a vector space V. Then for each $x \in V$ there exists unique scalars $c_1, \ldots, c_n$ such that $x = c_1 b_1 + \cdots + c_n b_n$.

1. Prove the theorem.

Definition 1. *Suppose* $\mathcal{B} = \{b_1, \ldots, b_n\}$ *is a basis for a vector space* V. *The* **coordinates of** x **relative to** $\mathcal{B}$ *are the weights* $c_1, \ldots, c_n$ *such that* $x = c_1 b_1 + \ldots c_n b_n$. *We define the* **coordinate mapping** *from* V *to* $\mathbb{R}^n$ *by* $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$.

2. Let $V = \mathbb{R}^3$ with basis $\mathcal{E} = \{e_1, e_2, e_3\}$. Let $x = 3e_1 + 4e_2 + 5e_3$, then find $[x]_{\mathcal{E}}$. Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$. Find $[x]_{\mathcal{B}}$.

3. Let $b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then $\mathcal{B} = \{b_1, b_2\}$ is a basis for $\mathbb{R}^2$. Let $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Find $[x]_{\mathcal{B}}$.

Definition 2. *Let $\mathcal{B} = \{b_1, \ldots, b_n\}$, then define $P_{\mathcal{B}} = [b_1 \ \ldots \ b_n]$. We call $P_{\mathcal{B}}$ the* **change of coordinates matrix**.

Note $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ and $[x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x$.

Theorem: Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ be a basis for a vector space V. Then the coordinate mapping $x \mapsto [x]_{\mathcal{B}}$ is a one-to-one, onto, linear transformation from V to $\mathbb{R}^n$.

4. Prove the theorem.

Definition 3. *Let $T : V \to W$ be a linear map of vector spaces. If T is one-to-one and onto, then T is a* **vector space isomorphism**, *and we say V and W are isomorphic. We write $V \cong W$.*

5. Show that $\mathbb{P}_3$ and $\mathbb{R}^4$ are isomorphic.

Theorem: If a vector space V has a basis $\mathcal{B} = \{b_1, \ldots, b_n\}$, then any set in V containing more than n vectors is linearly dependent.

6. Prove the theorem.

Cor: If V is a vector space with basis $\mathcal{B} = \{b_1, \ldots, b_n\}$, then each set of linearly independent vectors has at most n elements.

Theorem: If a vector space V has a basis with n elements, then every basis of V must have exactly n vectors.

7. Prove the theorem.

Definition 4. *If V is spanned by a finite set, then V is called* **finite dimensional,** *and the* **dimension** *of V, written $dim(V)$, is the number of vectors in a basis of V. Define $dim(\{0_V\}) = 0$. If V is not spanned by any finite set, then V is called* **infinite dimensional**.

Coordinates – Answers / Solutions

1. *Proof.* Since $\mathcal{B}$ is a basis, we know there exists scalars $c_1, \ldots, c_n \in \mathbb{R}$ such that

$$x = c_1 b_1 + \cdots + c_n b_n.$$

In order to prove the scalars are unique, we suppose $d_1, \ldots, d_n$ are scalars such that $x = d_1 b_1 + \cdots + d_n b_n$. Now subtracting the two representations of x, we get

$$0 = x - x = (c_1 b_1 + \cdots + c_n b_n) - (d_1 b_1 + \cdots + d_n b_n).$$

Using the vector space axioms, we rearrange the terms and factor to get

$$0 = (c_1 - d_1) b_1 + \cdots + (c_n - d_n) b_n.$$

Since $\mathcal{B}$ is a basis, we know $b_1, \ldots, b_n$ are linearly independent, and hence

$$c_1 - d_1 = 0, \ldots, c_n - d_n = 0.$$

Thus $c_1 = d_1, \ldots, c_n = d_n$. Thus the scalars are unique. $\square$

2. $[x]_{\mathcal{E}} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$. In order to find $[x]_{\mathcal{B}}$ we want to find scalars c_1, c_2, c_3 such that $x = c_1 b_1 + c_2 b_2 + c_3 b_3$. Thus we form the augmented matrix $[b_1 \ b_2 \ b_3 \ |x]$ and solve

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 5 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 5 \end{array}\right).$$

Thus $x = (-1)b_1 + (-1)b_2 + 5b_3$, and hence $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ -1 \\ 5 \end{bmatrix}$.

3. Again we form the augmented matrix corresponding to the vector equation $x = c_1 b_1 + c_2 b_2$: $\left(\begin{array}{cc|c} 1 & 2 & 1 \\ 1 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 1 \end{array}\right)$. Thus $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

4. *Proof.* Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ be a basis for the vector space V. Let $T : V \to \mathbb{R}^n$ be given by $T(x) = [x]_{\mathcal{B}}$.

Subclaim 1: T is linear.

Let $x, y \in V$. Since $\mathcal{B}$ is a basis, there exists (unique) scalars $c_1, \ldots, c_n, d_1, \ldots, d_n$ such that

$$x = c_1 b_1 + \cdots + c_n b_n,$$

and

$$y = d_1 b_1 + \cdots + d_n b_n.$$

Thus using the vector space axioms, we get

$$[x+y]_{\mathcal{B}} = [(c_1 + d_1) b_1 + \cdots + (c_n + d_n) b_n]_{\mathcal{B}} = \begin{bmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = [x]_{\mathcal{B}} + [y]_{\mathcal{B}}.$$

Let $c \in \mathbb{R}$, then $[cx]_{\mathcal{B}} = [cc_1 b_1 + \cdots + cc_n b_n]_{\mathcal{B}} = \begin{bmatrix} cc_1 \\ \vdots \\ cc_n \end{bmatrix} = c[x]_{\mathcal{B}}$.

Thus T is linear.

Subclaim 2: T is one-to-one (injective).

Let $x, y \in V$ such that $T(x) = T(y)$. By the first theorem today, the coordinates are unique, so $x = y$.

Subclaim 3: T is onto (surjective).

Let $f = \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix} \in \mathbb{R}^n$, then $f_1 b_1 + \cdots + f_n b_n \in V$ and $T(f_1 b_1 + \cdots + f_n b_n) = f$. Thus T is surjective. $\square$

5. Let $\mathcal{B} = \{1, t, t^2, t^3\}$ be the standard basis on $\mathbb{P}_3$. By the previous theorem the coordinate mapping with respect to basis $\mathcal{B}$ is a vector space isomorphism.

6. *Proof.* Let $a_1, \ldots, a_p \in V$ with $p > n$, then $[a_1]_{\mathcal{B}}, \ldots, [a_p]_{\mathcal{B}} \in \mathbb{R}^n$. We have more than n vectors in $\mathbb{R}^n$, so we proved they must be linearly dependent. Thus there exists $c_1, \ldots, c_p \in \mathbb{R}$, not all zero, such that

$$0 = c_1[a_1]_{\mathcal{B}} + \cdots + c_p[a_p]_{\mathcal{B}}.$$

Since the coordinate mapping is linear, we get

$$0 = [c_1 a_1 + \cdots + c_p a_p]_{\mathcal{B}}.$$

Since the coordinate mapping is injective, we get $0 = c_1 a_1 + \cdots + c_p a_p$ where not all $c_1, \ldots, c_p$ are zero. Thus the vectors are linearly dependent in V. $\square$

7. *Proof.* Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ be a basis for V and let $\mathcal{C} = \{c_1, \ldots, c_m\}$ be a basis for V. Since $\mathcal{B}$ is a set of linearly independent vectors in V, the previous theorem tells us that $n \leq m$. Since $\mathcal{C}$ is a set of linearly independent vectors in V, the previous theorem tell us that $m \leq n$. Thus $n = m$. $\square$